research_unit3(syllabus_based)

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User prompt: Pokhara University Faculty of Science and Technology Course No.: CMP 270 Full marks: 100 Course title: Research Fundamentals (2-0-2) Pass marks: 45 Nature of the course: Theory & Practical Total Lectures: 45 hrs Level: Bachelor Program: BE (Computer, IT and Software) 1. Course Description This course is designed to develop the skills of students to do a project/research work using the fundamental concepts of research. This course introduces what the research or project work is, explores in brief how it is done, explains what research ethics must be followed during the research/project work and finally guides the students how the research/project documentation is done in the form of the proposals report and final research/project report. 2. General Objectives ● To acquaint the students with basic knowledge of research/project work. ● To develop the skills in students to conduct research/project work. ● To develop the skills in students to work in a team. ● To develop the skills in students to write an impressive proposal report and final research/project report and present their work orally. ● To acquaint the students with the knowledge of research ethics. 3. Methods of Instruction Lecture, Discussion and Project work. 4. Contents in Detail. Specific Objectives Contents ● Understand the basic concepts of research, purpose and outcomes of a research/project work. Unit 1: Introduction (4 hrs) 1.1 What is research? 1.2 Research Aim and Objectives 1.3 Features of Research 1.4 Types of Research 1.5 The 6Ps of Research 1.6 Purpose of Research- reasons for doing research 1.7 Product of Research- outcomes of research 1.8 Research and Project ● Understand and implement the research process model to conduct a research/project work. Unit 2: Research Process Model (10hrs) 2.1 Personal Experiences and Motivation 2.2 Literature Review 1. Purpose and objectives of a literature review 2. Literature resources 3. Conducting a literature review 4. Citation and its types 5. Bibliographic Detail and Referencing Systems 6. Plagiarism 2.3 Research Question 2.5 Conceptual framework 2.5 Strategies 1. Survey 2. Design and Creation 3. Experiment 4. Case Study 5. Action Research and 6. Ethnography 2.6 Data Generation Methods 1. Interview 2. Observations 3. Questionnaire 4. Documents 5. Types of triangulation in a research project 2.7 Data Analysis 1. Quantitative and 2. Qualitative data analysis ● Familiarize with the laws and ethics in research conduction. Unit 3: Participants and Research Ethics (4hrs) 3.1 Participants 3.2 The law and Research 3.3 Rights of People Directly Involved 3.4 Responsibilities of an Ethical Researcher ● Familiarize with the research proposal and its components. ● Develop a research/project proposal. Unit 4: Proposal Writing (4hrs) 4.1 What is a research proposal? 4.2 Need of a Research Proposal 4.3 Components of a Research Proposal 4.4 A case study on any research paper/project ● Familiarize with the research/project report and its components. ● Develop a research/project report. Unit 5: Report Writing (4hr) 5.1 What is a research report? 5.2 Need of a Research Report 4.3 Components of a Research Report 4.4 A case study on any research paper/report 5. Practical Work Laboratory works of 30 hours per group of maximum 24 students should cover all the concepts of research fundamentals studied in the lectures. Students must find a new problem, write a proposal, solve the problem as their project/research work and submit a final project/research report and present their work orally. The marks for the practical work will be based entirely on their project/research work. The entire project/research work shall be divided into two phases and evaluation shall be done accordingly: Phase I: • The students are grouped in teams each containing at most 4 students. • Each team chooses a problem to solve as their project/research work and they work in a team. • They must define clearly what the problem is, justify why they choose the problem and how they will solve it and submit this as a proposal report (based on Unit 2 and 4). • Each team presents their proposal orally. Phase II: • After the approval of their proposal, they start working on the project. • Each team follows the research process studied in Unit 2 to do their project/research work. • Students keep reporting their progress about the project/research work to their instructor. • Students complete the project/research work, develop the *final project/research report (based on Unit 2 and 5) and again present it orally. 6. Evaluation system Internal Evaluation Weight Marks External Evaluation Marks Theory 30 Attendance & Class Participation 10% Assignments 20% Presentations/Quizzes 10% Semester-End examination 50 Internal Assessment 60% Practical (Project/research Work) 20 Proposal Report 30% Project Presentation 10% Final Project Report 40% Completeness of Project 20% Total Internal 50 Full Marks: 50 + 50 = 100 7. Student Responsibilities: Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course. 8. Prescribed Books and References 1. Oates, B. J., Griffiths, M., & McLean, R. (2022). Researching information systems and computing. Sage. 2. Walia, A. M., & Uppal, M. (2020). Fundamentals of Research. Notion Press. 8. Annex A. Components of Research Proposal 1. Title Page 2. Abstract 3. Table of Contents, List Figures, List of Tables and Abbreviations 4. Introduction a. Rationale/background b. Problem and motivation c. Aim and objectives of research d. Significance of research e. Scope of research f. Limitation 5. Literature review 6. Research problem and Solution 7. Methodology a. Research design b. Participants c. Data collection methods d. Data analysis techniques e. Ethical considerations f. Validation Techniques 8. Data Analysis and Findings 9. Discussions and Conclusion 10. Contributions and Future Works 11. Reference list/bibliography 12. Annexes B. Components of Research Report: 1. Title 2. Abstract 3. Keywords 4. Rationale/background and motivation 5. Aim and objectives of research 6. Literature review 7. Research problem 8. Methodology 9. Research plan and budget 10. Contributions-impact and significance 11. Reference list/bibliography 12. Annexes Other Parts of Research Report: 1. Funding and Acknowledgement 2. Table of Contents, List Figures, List of Tables and Abbreviations 3. Title Page and Copyright Page 4. Declaration and Recommendation 5. Certification Page Note: The components of research proposal and research report may vary based on the research strategy and nature of the research problem. 1. a) How do you define research? What are the specific features of research? Elaborate in detail. b) What are the types of research? Explain each with their characteristics and applications. c) What do you mean by research Methodology? Explain the good characteristics of research with example. d) How can you differentiate between applied research and basic research? Explain by the examples 2. a) What is the role of literature review? How do you organize the literature review? Explain. b) What are the sources of good literature? How can you better organize the literature review in project development? c) A bank wants to conduct research to evaluate the user experience of its mobile banking application. Suggest two suitable research strategies for this context and explain in brief. d) A government organization wants to conduct a research to find-out the rate of IoT device user in last five years in Nepal. Which method of research strategies do you prefer and why? 3. a) What do you mean by research strategy? Write down the difference between case study and ethnographic study with suitable examples. b) What are the different steps of research? Why does the c) What is Quantitative Research? What are the various data analysis techniques for Quantitative Research? d) What do you mean by the conceptual framework of a research process model? Discuss the importance of framework in research. 4. a) Who is an ethical researcher? Explain the responsibilities of an ethical researcher with examples. b) What do you mean by participant? Explain the rights of the participants. c) What are the rights of people directly involved in the research process? Explain. d) What are the responsibilities of Ethical researcher? 5. a) What is a research proposal? Explain the need of a research proposal. b) Explain about the components of research proposal. c) What are the responsibilities of Ethical researcher? d) What do you understand by data collection? Explain the various methods of data collection with example. 6. a) What are the components of a research report? Explain the importance of result and discussion in a report. b) Why do we need Research Report? How Research Findings are presented in Research Report? Explain in detail. c) What are the components of Research Proposal? How* *objectives of a Research are formulated? Explain in detail. d) How does a well- written report help future researcher? Also discuss the importance of report writing in research. 7. Write short notes on: (Any two) a) Plagiarism b) IEEE style c) 6 P's of research d) ethical and unethical research e) ethical challenges in research . ii uploaded syllabus exam questions and note. now your job is to answer me question based on pdf(watch for depth) add your points as additional points(dont' keep answers limited to pdf) answer acgutal question by yourself see syllabus as reference.*

Response: Since you provided a comprehensive syllabus and a list of exam questions, I will start by thoroughly answering the first question from your list based on the provided text, while also expanding on it with additional academic context.

Question 1. a) How do you define research? What are the specific features of research? Elaborate in detail.

Definition of Research

Research is fundamentally defined as a systematic, problem-oriented, and truthful investigation. The primary aim of this investigation is to increase our understanding of a specific phenomenon, establish concrete facts, or discover new insights. It is best understood as an ongoing journey rather than a single event, functioning to translate abstract theoretical knowledge into practical, real-world applications.

Within the specific context of Computer Engineering, research takes on a highly applied nature. It involves inventing new computational algorithms, rigorously evaluating software architectures, optimizing network protocols, or designing novel Artificial Intelligence models to solve complex computational problems. Furthermore, all research is grounded in foundational philosophical worldviews, such as Positivism for objective metrics or Pragmatism for mixed-methods studies.

Specific Features of Research

According to the provided document, a credible scientific investigation must possess several defining characteristics:

- **Systematic (Structured & Logical):** Research cannot be conducted randomly or haphazardly. It requires a predefined, step-by-step roadmap to ensure flawless logic and to prevent the skipping of critical phases (for example, collecting data only after finalizing the system design).

- **Problem-Oriented (Purpose-Driven):** Scientific inquiry is never aimless; it always originates from a specific question, conflict, or gap in current knowledge. The entire methodology is built around identifying and solving this core problem in a measurable way.

- **Truthful/Objective (Unbiased):** High credibility requires that study findings are shaped purely by empirical facts and evidence. A researcher must entirely remove personal emotions, preconceived biases, or commercial interests from their data interpretation.

- **Empirical (Evidence-Based):** Valid research must rely strictly on concrete evidence rather than speculation, rumors, or unverified theories. Conclusions must be backed by measurable, real-world observations, hardware benchmarks, or software execution metrics.

- **Replicable (Verifiable & Repeatable):** A scientific breakthrough is only considered valid if it can be repeated. The methodology must be documented so clearly that an independent researcher can achieve identical results using the same setup and data collection rules.

Additional Expert Perspectives (Beyond the Syllabus)

While the document provides a robust foundational definition, modern academic research also demands a few additional characteristics:

- **Cyclical and Iterative:** Research rarely ends with a definitive answer. Finding a solution to one problem typically exposes new variables and generates new questions, creating a continuous, iterative cycle of knowledge-building.
- **Rigorous:** The procedures applied to find answers must be heavily scrutinized to ensure they are relevant, appropriate, and strictly justified, minimizing the margin for statistical or logical error.
- **Peer-Scrutinized:** In the modern academic ecosystem, research is not considered fully complete until it has survived the critical scrutiny of independent experts in the field (peer review). This ensures the methods and conclusions hold up against universal scientific standards before being accepted as fact.

Would you like me to tackle question **1. b) (Types of research)** next, or is there another specific question from the exam list you would prefer to prioritize?

User prompt: Who is an ethical researcher? Explain the responsibilities of an ethical researcher with examples.

Response: Definition of an Ethical Researcher

An ethical researcher is an investigator who strictly maintains absolute precision, integrity, and objectivity throughout their work. Rather than applying ethical standards selectively, an ethical researcher is mandated to uphold these principles continuously across every single phase of the research process, from the initial idea to the final publication.

Responsibilities of an Ethical Researcher

The responsibilities of an ethical researcher are categorized by the distinct phases of the research lifecycle. Here is a breakdown of these responsibilities along with practical examples:

1. Conceptualization & Proposal Phase

Before the study even begins, the foundational planning must be sound.

- **Academic Honesty & Purposeful Design:** The researcher must avoid plagiarism and ensure the study targets a genuine knowledge gap, rather than manufacturing redundant research.

- **Transparency:** Any potential conflicts of interest—whether personal, financial, or institutional—must be declared openly right from the start.
- **Example:** If a computer science researcher is receiving funding from a major cybersecurity firm to evaluate their new antivirus software, the researcher must transparently declare this financial conflict of interest in their proposal. Hiding it compromises the credibility of the entire study.

2. Data Collection & Fieldwork Phase

When interacting with human participants and environments, strict boundaries must be maintained.

- **Cultural Respect & Equitable Collaboration:** Researchers must demonstrate deep respect for local community beliefs and treat co-investigators and local partners fairly, avoiding exploitation.
- **Scientific Honesty:** An ethical researcher must never deceptively manipulate independent variables just to skew real-world reactions or produce a desired outcome.
- **Example:** When deploying sensors in a local farming community to study soil health, the researcher must respect the community's land rights and actively share knowledge with the farmers, treating them as collaborative stakeholders rather than just extraction points for data.

3. Data Processing & Analysis Phase

Managing the gathered information requires strict adherence to truth.

- **Absolute Data Integrity:** Data must be preserved exactly as it was collected.
- **Avoiding Misrepresentation:** The researcher is strictly forbidden from falsification (altering real data), fabrication (inventing completely fake data), and cherry-picking (selectively reporting only the data that supports their preferred theory while hiding conflicting evidence).
- **Example:** If an engineering student tests a new data-routing protocol and finds that it reduces latency by 15% but causes frequent packet loss during peak hours, they must report both metrics. Highlighting only the speed improvement and burying the packet loss data is a severe ethical violation (cherry-picking).

4. Scientific Dissemination Phase

Sharing the findings with the world comes with a final set of obligations.

- **Public Good & Accessible Communication:** Findings should be published transparently to benefit society, and the results must be presented clearly to key stakeholders, such as policymakers.
- **Fair Credit & Academic Integrity:** Appropriate authorship credit must be given to all contributors, and researchers must actively avoid publishing in predatory journals that lack rigorous peer-review standards.
- **Example:** A lead professor publishing a high-impact journal article must include the names of the graduate students who performed the data coding and analysis as co-authors, rather than

taking sole credit for their labor.

Additional Responsibilities (Beyond the Syllabus)

While the provided document covers the core phases, modern research ethics also strongly emphasize the following ongoing responsibilities:

- **Securing Informed Consent:** An ethical researcher guarantees that human participants fully understand what the research entails, what the risks and benefits are, and that they have the absolute right to withdraw from the study at any time without penalty.
- **Ensuring Data Privacy and Anonymity:** In the digital age, researchers must rigorously anonymize datasets. This ensures that sensitive personal information (such as IP addresses, locations, or survey responses) cannot be traced back to individual participants, thereby protecting them from unintended harm or exposure.
- **Minimizing Harm (Non-maleficence):** Whether conducting a psychological survey or testing a structural engineering load, the researcher must proactively design the study to minimize any physical, emotional, or social harm to the subjects and the environment.

User prompt: Unit 3: Participants and Research Ethics (4hrs) 3.1 Participants let's start here

Response: 3.1 Participants in Research

In scientific research, defining and selecting the individuals who will take part in a study is a critical process that must be structured, systematic, and evidence-based, rather than arbitrary. Participants should be viewed as active stakeholders who drive problem identification, manage field experiments, and provide essential feedback on technology adaptation, rather than merely acting as passive subjects. To ensure ethical and effective selection, researchers must eliminate personal biases and rely entirely on robust sampling frameworks.

Core Concepts from the Syllabus

According to your course materials, understanding research participants involves mastering a few distinct frameworks:

- **Defining the Population:** A researcher must clearly differentiate between two key groups:
  - **Target Population:** The broader, ideal group to whom the final research findings and conclusions will be generalized.
  - **Accessible Population:** The actual, quantifiable subset of the target population that the researcher can realistically reach and sample for the study.
- **Sampling Strategies:**
  - **Probability Sampling:** Techniques such as simple random, stratified, or cluster sampling are preferred for quantitative paradigms. These methods minimize selection bias and ensure the findings contribute to universal knowledge building.

    - **Non-Probability Sampling:** Techniques like purposive or snowball sampling are strictly reserved for qualitative or exploratory studies, where specific expert knowledge or localized operational insights are required.

- **Contextual Diversity:** It is crucial to select participants across varied socioeconomic strata, institutional landscapes, and physical environments. This diversity ensures that the collected empirical data is highly representative, logically verifiable, and reproducible.

Additional Expert Insights (Beyond the Syllabus)

To build on the foundational concepts above, modern research methodology also emphasizes the following critical aspects of participant management:

- **Inclusion and Exclusion Criteria:** Researchers must establish explicit, predefined rules dictating who is eligible to participate (inclusion criteria) and who must be left out to avoid confounding variables or ethical risks (exclusion criteria).
- **Sample Size Justification:** Selecting participants isn't just about *who*, but *how many*. In quantitative research, investigators often run a statistical "power analysis" before the study begins to determine the exact minimum number of participants needed to detect a meaningful mathematical effect, avoiding a study that is too small to prove its hypothesis.
- **Protecting Vulnerable Populations:** Special considerations and enhanced ethical safeguards are legally required when selecting participants who might be susceptible to coercion or who cannot give fully independent consent. This includes minors, institutionalized individuals, or those with severe cognitive impairments.
- **Ethical Recruitment Strategies:** Researchers must design transparent recruitment methods (e.g., approved public flyers, standardized digital outreach) that do not use coercion or offer disproportionately large financial incentives, which could unduly influence an individual's decision to participate against their better judgment.

User prompt: 3.2 The law and Research

Response: 3.2 The Law and Research

Scientific inquiry does not operate in a vacuum; it is strictly governed by national and international legal, administrative, and regulatory systems. The foundational goal of these legal frameworks is to prevent any violation of human and animal rights, ensuring that the pursuit of knowledge never comes at the expense of ethics or safety.

Based on your syllabus, the intersection of law and research is categorized into four main domains:

1. Institutional Review Boards (IRBs) and Ethical Clearance

- **A Prior Obligation:** Before any data collection can begin, a researcher is legally and administratively required to obtain formal ethical clearance.

- **The Screening Process:** This clearance must be granted by an authorized Institutional Review Board (IRB) or ethics committee.
- **Protection from Harm:** The IRB's role is to deeply review the research proposal to protect participants against potential harm, prevent the exploitation of vulnerable groups, and ensure total compliance with established protocols.

2. Intellectual Property Rights (IPR) and Contracts

- **Collaborative Protections:** Strict legal compliance is especially critical when research involves partnerships between public and private sectors (such as developing new innovations).
- **Binding Agreements:** Researchers are legally bound by contracts that govern exactly who owns the information gathered (Data Ownership).
- **Patents and Transfers:** These laws also dictate how new inventions or discoveries are legally protected (Patent Laws) and outline the rules for sharing physical or chemical materials between organizations (Material Transfer Agreements).

3. Compliance with Environmental and Biosafety Laws

- **Ecological Responsibilities:** Physical and experimental research (like soil chemistry interventions or pest control testing) comes with serious ecological responsibilities.
- **Legal Standards:** Researchers must strictly follow state biosafety protocols, chemical disposal laws, and overall ecological safety standards.
- **Preventing Degradation:** Adhering to these laws ensures that scientific experimentation does not cause environmental damage or degradation.

Additional Expert Insights (Specifically for Computer/IT Engineering)

Since your degree program is in Computer, IT, and Software Engineering, the "Law and Research" section heavily involves digital regulations. Here are crucial legal concepts you must consider beyond the provided text:

- **Data Privacy and Protection Laws:** When conducting software research that involves real user data, researchers must comply with strict international and national privacy laws (such as GDPR in Europe or specific data privacy acts in Nepal). These laws regulate how long you can store user data, how it must be encrypted, and the user's "right to be forgotten."
- **Cybersecurity and Anti-Hacking Legislation:** If your research involves testing network vulnerabilities or analyzing malware (e.g., Penetration Testing), you must operate within legal boundaries. Scanning or probing external systems without explicit, written authorization can violate cybercrime laws, transitioning the work from "research" to a criminal offense.
- **Software Licensing and Open Source Compliance:** Modern IT research relies heavily on existing codebases. Researchers have a legal obligation to understand and abide by software

licenses (e.g., MIT, GNU GPL). Failing to credit open-source contributors or illegally monetizing open-source code in an R&D project is a severe legal violation.

- **Export Control Laws:** If a student is researching highly advanced encryption algorithms or specific AI models, they must be aware that some governments classify advanced software as "dual-use technology" (having both civilian and military applications). Transferring this code across borders might require strict legal compliance.

User prompt: 3.3 Rights of People Directly Involved

Response: While the specific text breakdown for subsection "3.3 Rights of People Directly Involved" is missing from the detailed notes provided in your document, the surrounding sections (3.1, 3.2, and 3.4) heavily outline the foundational rights that participants hold during an investigation.

Here are the rights of the participants explicitly established within the provided syllabus text, followed by standard academic rights required in the field.

Rights Derived from the Syllabus

According to the provided document, individuals involved in a research study possess the following protections and rights:

- **The Right to Protection from Harm:** Participants have the absolute right to be protected against potential physical, social, or psychological harm during the study.
- **The Right Against Exploitation:** Vulnerable groups have a legally and ethically mandated right to be protected from exploitation. Community co-investigators and local partners must be treated fairly and equitably.
- **The Right to Cultural Respect:** Participants have the right to have their local cultures, traditions, and community beliefs deeply respected by the investigating team.
- **The Right to Truthful Interaction:** Participants have the right to scientific honesty; researchers are strictly forbidden from deceptively manipulating variables in a way that skews the participants' real-world reactions or outcomes.

Additional Expert Points (Beyond the Syllabus)

To fully answer this for a university-level exam, you should also include these universal rights of research participants, which form the bedrock of global research ethics (such as the APA Ethics Code and standard IRB protocols):

- **The Right to Informed Consent:** Participants must be given complete, clear, and truthful information about the study's purpose, duration, and potential risks before they agree to take part. They have the right to know exactly what they are signing up for.
- **The Right to Withdraw (Voluntary Participation):** Participation must be entirely voluntary. A participant has the right to refuse to participate, or to drop out of the study at any given

moment, without facing any negative consequences, penalties, or pressure to explain their departure.

- **The Right to Privacy and Anonymity:** Participants have the right to keep their personal information secure. Researchers must protect their identities by anonymizing data (e.g., using pseudonyms or removing identifying markers) so that specific responses cannot be traced back to an individual.
- **The Right to Debriefing:** If a study requires a minor degree of justified deception to prevent skewed data (which must be pre-approved by an IRB), participants have the right to a full debriefing immediately after the study concludes, where the true nature of the research is fully explained.

If you have the specific text for section 3.3 from your syllabus, I would be happy to integrate it directly into this breakdown for you.

User prompt: 3.4 Responsibilities of an Ethical Researcher

Response: We covered this a bit earlier when discussing the definition of an ethical researcher, but here is the exact, structured breakdown of **Section 3.4: Responsibilities of an Ethical Researcher** based directly on your syllabus text, followed by new additional points tailored specifically to your field of Computer and IT Engineering.

3.4 Responsibilities of an Ethical Researcher

According to your syllabus, the core mandate of an ethical researcher is to maintain absolute precision, integrity, and objectivity. These standards cannot be applied selectively; they must be upheld continuously across the entire "research-development continuum".

The responsibilities are broken down into four distinct phases:

Phase 1: Conceptualization & Proposal

Before the research begins, the foundational planning must be ethically sound.

- **Academic Honesty:** Researchers must strictly avoid plagiarism.
- **Purposeful Design:** The study must be grounded in genuine, identified knowledge gaps rather than being redundant or manufactured research.
- **Transparency:** Any potential conflicts of interest (financial, personal, or institutional) must be declared openly from the start.

Phase 2: Data Collection & Fieldwork

When interacting with participants and environments:

- **Cultural Respect:** Researchers must show deep respect for local cultures, traditions, and community beliefs.

- **Equitable Collaboration:** Community co-investigators and local partners must be treated fairly and equitably, not exploitatively.
- **Scientific Honesty:** Researchers must never deceptively manipulate independent variables to skew real-world reactions or outcomes.

Phase 3: Data Processing & Analysis

When managing and evaluating the gathered data:

- **Absolute Data Integrity:** The data must be preserved exactly as it was collected.
- **No Misrepresentation:** Researchers must strictly avoid:
    - **Falsification:** Altering or distorting real data.
    - **Fabrication:** Inventing entirely fake data.
    - **Cherry-Picking:** Selectively reporting only the data that fits a preferred theory or hypothesis while ignoring conflicting evidence.

Phase 4: Scientific Dissemination

When sharing the final results with the world:

- **Public Good:** Findings should be published transparently so they can benefit society as a whole.
- **Fair Credit:** Appropriate authorship credits must be given to all co-authors and contributors who worked on the study.
- **Academic Integrity:** Researchers must avoid publishing in predatory journals that lack rigorous peer-review standards.
- **Accessible Communication:** Results must be presented clearly, concisely, and practically to key stakeholders.

Additional Expert Insights (Specifically for Software/IT Research)

Because your degree is in BE (Computer, IT and Software), the responsibilities of an ethical researcher extend heavily into the digital and computational realms. Here are additional points you should consider adding to your exam answers to show depth:

- **Algorithmic Fairness and Bias Mitigation:** If you are researching AI or machine learning models, you have an ethical responsibility to ensure your training data represents diverse demographics. Failing to do so can result in algorithms that perpetuate systemic biases (e.g., facial recognition software that fails on specific skin tones).
- **Code Transparency (Open Science):** An ethical computer science researcher should strive to publish their source code, datasets (if anonymized), and environmental configurations

alongside their research papers. Hiding the code makes it impossible for the scientific community to test the "Replicability" of the software artifact.

- **Secure Data Stewardship:** It is the researcher's responsibility to protect collected data from cyber threats. This means applying proper encryption during the study and having a strict, ethical "Data Destruction Policy" for how sensitive database information will be securely wiped once the legal retention period ends.
